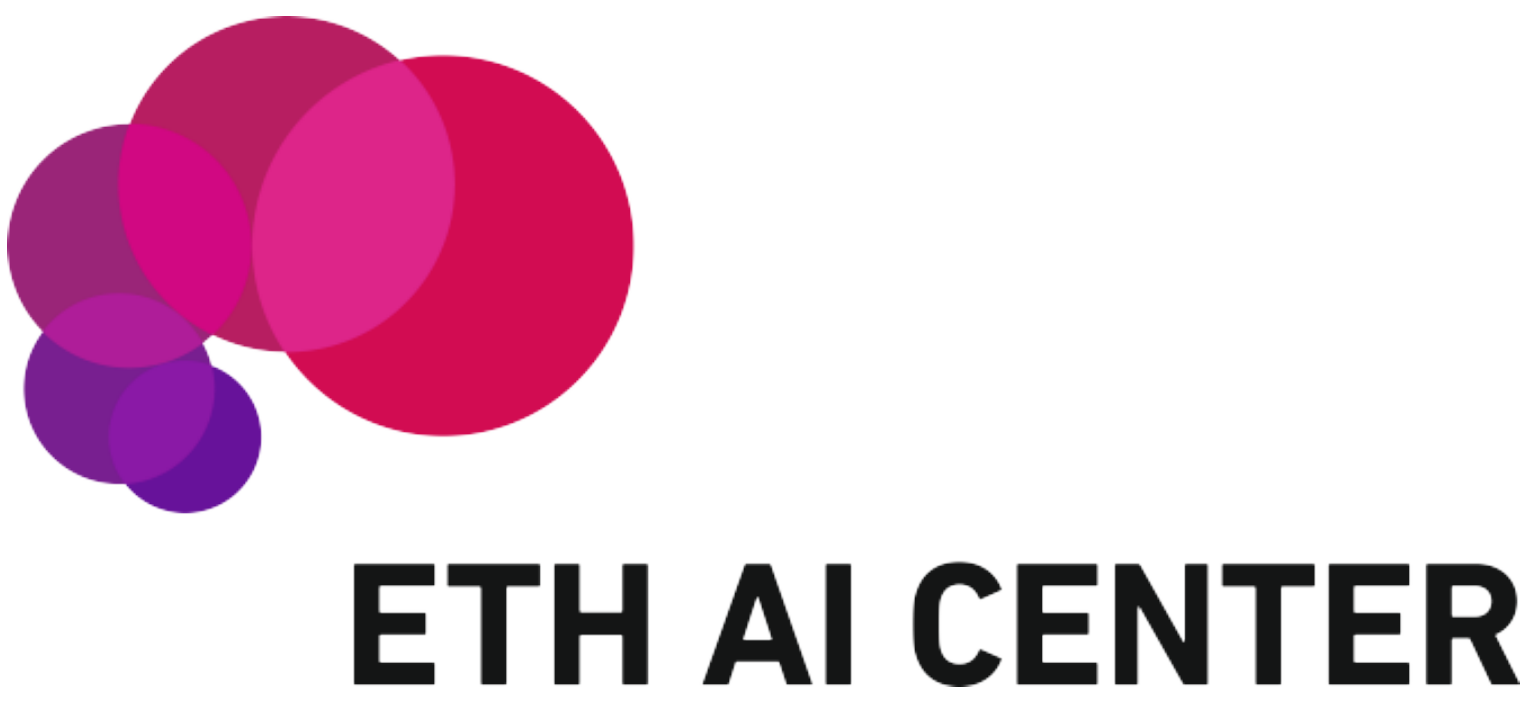




Slack-integrated AI Knowledge Agent for ETH AI Center.

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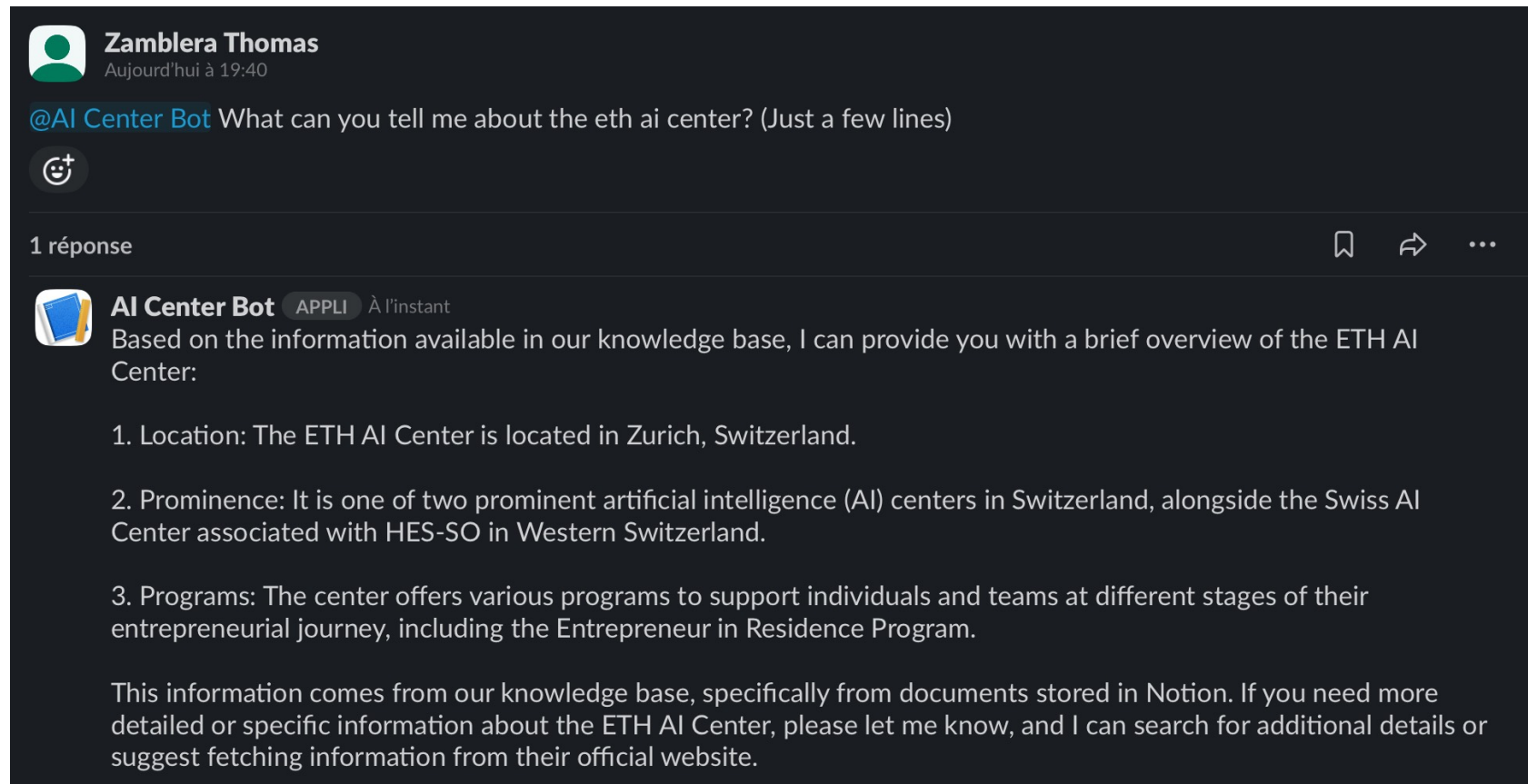


Figure 1: Example of interaction with the chatbot in a Slack channel

1. Problem Statement

The ETH AI Center generates and manages a large amount of institutional knowledge across various platforms — Google Drive, Notion, Airtable, and internal websites. However, this information is often fragmented and not easily searchable by staff and researchers.

Our goal was to design and prototype a Slack-integrated AI chatbot capable of answering queries about ETH AI Center operations, research, and documentation using a centralized knowledge base.

2. Data

To serve relevant and up-to-date answers, the chatbot relies on a diverse set of data sources, both public and private:

- **Google Drive:** Internal documents, meeting notes, proposals, and administrative material shared within the ETH AI Center.
- **Notion:** Project documentation, onboarding content, and structured knowledge pages used by staff and fellows.
- **Airtable:** Operational trackers and databases including lists of fellows
- **Internal Wiki:** Centrally maintained documentation on processes, tools, FAQs, and onboarding guides specific to the ETH AI Center
- **ETH Website & News:** Crawled pages from `ai.ethz.ch` and the official ETH news site, contains latest public announcements
- **Slack Messages:** Contextual threads and frequently asked questions extracted from internal Slack channels

All sources were indexed into a centralized vector database, enabling semantic search and grounding of language model responses.

3. Methods

Our assistant is built around a hybrid architecture for dynamic and static content retrieval, integrating Claude via the MCP framework.

System Architecture

The system comprises three main components:

- **Claude interactions:** Anthropic’s Claude uses MCP to access tools like search, document fetch, web crawling, and Slack. Bot interactions are stored in a database as separate conversations, which serve as contextual memory
- **Knowledge Sources:** Dynamic content (e.g., Slack threads) is fetched on demand via tools, while static content (e.g., Notion, Airtable, internal wikis) is indexed into a centralized vector database daily
- **MCP Tool Suite:** Custom tools are implemented for:
 - `slack_get_thread_replies`, `slack_get_channel_history`
 - `search`, `fetch_document` for semantic document retrieval
 - `web_fetch` for real-time web scraping

Query Handling Flow

When a user asks a question in Slack:

1. The bot logs the message and retrieves conversation history from (1)
2. A system prompt is generated with context and source citation rules
3. Claude uses reasoning and tools to:
 - Retrieve Slack context if needed
 - Search and fetch relevant documents from the vector DB
 - Access real-time data from URLs or Google Drive
4. The model composes a response with sources citations and either calls the `slack_reply_to_thread` tool

Knowledge Indexing

A dedicated sync pipeline runs frequently to:

- Crawl ETH AI Center sources (Google Drive, Notion, Airtable, internal wiki, `ai.ethz.ch`)
- Chunk and embed documents via a document processor
- Store semantic representations into a vector DB for efficient retrieval

Deployment

The bot runs currently on a VPS but will be deployed on a Kubernetes server. A custom Slack app handles mentions/DMs, deduplicates inputs, and communicates with the MCP server. It can also respond to emails sent to `eth.ai.assistant@gmail.com`.

4. Results

The assistant is fully functional, handling Slack and email queries using Claude and MCP. It retrieves and cites relevant information from multiple sources with contextual awareness. Early demos show strong potential for broader deployment.

5. Conclusions

We developed a Slack-integrated AI assistant tailored to the ETH AI Center, capable of retrieving information from diverse sources such as Slack, Google Drive, Notion, and public websites. By combining a semantic index for static content with real-time tool-based retrieval for dynamic queries, our MCP-based system balances scalability and freshness. Although testing was limited in scale, the architecture serves as a promising proof of concept for organization-level knowledge assistance.

6. Future Work

- **Scale testing with more real data:** Access to internal systems was delayed, so many sources were manually populated. Once compliance is approved, validating performance on real-world data will be crucial
- **User feedback loop:** Adding in-Slack feedback (e.g., rating responses) would help assess quality, guide improvements, and inform potential fine-tuning of retrieval and summarization
- **Query clustering and analytics:** Grouping similar queries and tracking recurring topics could support response monitoring and highlight unmet needs or content gaps

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Contributions

Ambroise: Slack MCP tool, general organisation/integration of codebase, database logic
Erges: Airtable MCP tool, Web crawler, general organisation/integration of codebase, Slack app and server deployment, Git repository management and documentation
Wesley: Notion scraper, Web crawler, research on Kubernetes, Git repository management and documentation
Thomas: Google-drive scraper, email integration, database logic, Slides and presentations
CC/AC, I. Schlag: Project supervision, conceptual guidance, and administrative coordination